

Faraz Iliaee, M.Sc.

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HIGHLIGHTS:

Results-driven **Analytical Chemist** with **3+ years of experience** in organic chemicals characterization and advanced analytical instrumentation. Demonstrated expertise in **high-resolution mass spectrometry**, **chromatographic techniques**, **method development**, **design of experiments (DoE)**, and data-driven investigation of complex chemical systems across environmental and biological matrices. Skilled in transforming complex analytical datasets into meaningful insights through **statistical modeling**, **chemometric analysis**, and computational tools, supporting characterization, risk evaluation, and scientific decision making. Proficient in **R**, **Python**, and analytical workflows for **targeted** and **non-targeted chemical analysis**.

QUALIFICATIONS:

ANALYTICAL TECHNIQUES: Ultra-High-Performance Liquid Chromatography (UHPLC), Liquid Chromatography-Mass Spectrometry (LC-MS; Orbitrap, Q-TOF, QqQ), Inductively Coupled Plasma Mass Spectrometry (ICP-MS), Gas Chromatography (GC), UV-Visible Spectroscopy, Fourier Transform Infrared Spectroscopy (FT-IR), Solid-Phase Microextraction (SPME), Passive Air Sampling (PUF Disks), Soxhlet Extraction, Rotary Evaporation (Rotovap), Nitrogen Evaporation, Karl Fischer Titration

COMPUTATIONAL TOOLS & SOFTWARE: R, Python, Power BI, SQL, ArcGIS, Gaussian, EPI Suite, Compound Discoverer, Xcalibur, MassHunter, OpenLab CDS, ChemDraw, LIMS, Microsoft Office

PROFESSIONAL COMPETENCIES: Instrument Troubleshooting, Problem Solving, Scientific Writing, Technical Presentations, Oral Communication, Cross-Functional Collaboration, Lean Six Sigma Practices

PROFESSIONAL EXPERIENCES:

RESEARCH ASSISTANT, CONCORDIA UNIVERSITY, MONTREAL, QUEBEC, CANADA, SEP 2024- SEP 2026

- Optimized **high-throughput analysis method** using **Xcalibur** for **300+ samples** supporting quantification of organic contaminants at **trace (ng/L) concentrations** across water, air, soil, and dust matrices.
- Developed **non-targeted screening workflows** using **Compound Discoverer** and **mzCloud** libraries to evaluate **>20,000 mass spectral features**, achieving **>99% reduction in dataset complexity** and confirming previously unreported transformation products.
- Applied **chemometric techniques**, **design of experiments (DoE) approaches**, and **data visualization tools** using **R**, **Python**, and **Power BI** to identify temporal trends and environmental fate relationships within complex analytical datasets.
- Conducted **prioritization** and **risk evaluation** of contaminants using **EPI Suite** and **CompTox Chemical Dashboard** to support **Canada's Chemicals Management Plan (CMP)** and communication of scientific findings to **government** and **public stakeholders**.
- Contributed to scientific publications and research funding through **two first-author manuscripts**, **five national and international conference presentations**, and **grant applications securing >\$20,000** in external funding.

TEACHING ASSISTANT & ACADEMIC OUTREACH LEAD, CONCORDIA UNIVERSITY, MONTREAL, QUEBEC, CANADA, SEP 2024- SEP 2026

- Instructed **10+ undergraduate laboratory and tutorial sections** spanning General, Organic, Analytical, and Physical Chemistry courses, mentoring **40+ students per semester** in experimental techniques, data analysis, and scientific reporting.
- Delivered **demonstrations** and **hands-on training** for analytical instrumentation, including **gas chromatography (GC)**, **rotary evaporation (Rotovap)**, **potentiometric titration systems**, and other equipment used in undergraduate chemistry curricula.
- Evaluated **laboratory reports** and **assignments** while providing individualized academic support through office hours and **mentoring sessions**, helping students improve **problem-solving skills** and achieve test score improvements of up to **30 percentage** points.
- Developed **partnerships** with industry, academic, and professional stakeholders, securing **sponsorship** support from organizations including **Waters** and **Fisher Scientific** for the 26th and 27th Annual Concordia **CBGRC Conferences**.

QUALITY CONTROL ANALYST (CO-OP), GOLRANG INDUSTRIAL GROUP, ZANJAN, ZANJAN, IRAN, SEP 2022- JAN 2024

- Performed routine and non-routine quality control testing in a **fast-paced GMP-compliant** environment, while delivering **analytical reports** and **technical documentation** within strict operational timelines.
- Performed testing of **personal care** and **household consumer products**, including **HPLC**, **ICP-MS**, **FT-IR**, and **Karl Fischer titration**, to support preservative analysis, heavy metal compliance, raw material qualification, and moisture determination.
- Executed **method verification** and **validation studies**, including accuracy, precision, recovery, linearity, LOD, and LOQ assessments, contributing to **optimization** of analytical procedures that **improved detection sensitivity** and **reduced analysis turnaround time** by approximately **5%**.
- Maintained **analytical records**, **test results**, and **batch documentation** within **LIMS** to support data integrity, traceability, audit readiness, and product release activities.
- Presented findings to **cross-functional teams** and **collaborated with R&D** groups to support product development and process optimization initiatives aligned with **Lean Six Sigma principles**.

EDUCATION:

M.Sc. Chemistry, Concordia University, Montreal, Quebec, Canada

2024-2026

(GPA: A), Advisor: Dr.Xianming Zhang

M.Sc. Thesis: Release, Transformation, and Environmental Occurrence of Rubber-Derived Chemicals in Aquatic Environments.

Safety & Laboratory Training: WHMIS, Hazardous Waste Disposal, Hazardous Materials Spill Response, Safe Storage of Hazardous Materials

B.Sc. Chemistry, University of Zanjan, Zanjan, Iran

2020-2024

(GPA: A), Advisor: Dr.Abbasali Zamani

B.Sc. Final Project: Determination of Losartan Potassium in the Presence of Hydrochlorothiazide Using Magnetic Solid-Phase Extraction and UV-Vis Spectroscopy in Wastewater Samples.
